Access beyond ownership: Stock Connect closures and price delay in Chinese stocks

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Highlights

• Stock Connect investors cannot trade on Hong Kong holidays.

• Prices of the stocks they hold then absorb index futures moves more slowly.

• The delay is absent on US holidays, when the channel stays open.

• A partial-adjustment model ties the delay to the capital that is shut out.

Abstract

Foreign investors trade Chinese A-shares mainly through Stock Connect, which shuts on Hong Kong holidays while the mainland market stays open. We use 45 such closures between 2017 and 2026 to separate foreign access from foreign ownership. In a partial-adjustment model, traders who act on market-wide news set how fast a stock's price absorbs it, so shutting out part of their capital raises the stock's price delay in proportion to the holdings of the excluded investors. On closure days, the price delay relative to index futures rises more for stocks with larger holdings through the channel. The rise appears with the closure, is at least as large when only Hong Kong is on holiday, and is absent on US market

holidays, when the channel is open. About a third of it remains after controls for size, index membership, analyst coverage, and exposure to Hong Kong prices, and most of that remains after controls for market activity on the day. Spreads do not widen.

Keywords: Foreign investors; Market access; Market segmentation; Stock Connect; Price delay

JEL classification: F21; F36; G14; G15

1. Introduction

International business research views foreign institutional investors as carriers of information and practices into emerging markets (Areneke et al., 2022; Kong et al., 2024). In stock markets, their trades speed up the response of local prices to global and market-wide news (Bae et al., 2012), and their ownership raises price efficiency across countries (Kacperczyk et al., 2021). In China, the Shanghai and Shenzhen-Hong Kong Stock Connect has opened the A-share market to foreign investors, and studies relate the opening to more informative prices, lower insider opportunism, and higher productivity (Lei and Lu, 2024; Liu et al., 2023; Yuan et al., 2023). These studies compare stocks or periods with and without foreign ownership. They cannot tell whether the benefits come from who owns the stocks or from the ability of foreign owners to trade at each moment, because ownership and access arrive together and stay together. The distinction matters for a segmented market (Kedar-Levy et al., 2025). If ownership is what counts, opening the door once is enough. If access is what counts, any interruption of the channel removes the benefit while the owners are still there.

Stock Connect lets us separate the two. Northbound trading follows the Hong Kong calendar, so on Hong Kong public holidays that are not mainland holidays, foreign investors who hold A-shares through the channel cannot trade, whereas the mainland market and its

index futures stay open. The owners are unchanged, but their access is switched off for a day or two on a calendar set months in advance. Consider a stock in which foreign investors hold 3% through the channel. When the CSI 300 index futures contract rises by 0.5% in one minute on a normal day, foreign investors are among those who buy the stock as the market news arrives, so much of the stock's response arrives within a minute or two. On a Hong Kong holiday, this part of the trading is missing. Our question is whether the stock's price then takes longer to absorb the futures move, and whether the delay is larger for stocks with larger holdings through the channel.

We frame the question with a partial-adjustment model in which the price of a stock closes a fixed fraction of the gap to its efficient value each minute, and this fraction rises with the capital of the traders who trade the stock on market-wide news. The model gives the price delay of Hou and Moskowitz (2005) in closed form. Delay depends only on the speed of adjustment, not on the stock's market exposure, the volatility of the market, or firm-specific news. If northbound holdings proxy for capital that can trade only when the channel is open, a closure lowers the speed of adjustment in proportion to the holdings, and the delay rises. The model also predicts that the effect requires the channel to close and that holdings of foreign investors with other access play no role.

We test these predictions with 45 closure episodes between 2017 and 2026. For each stock and day, we estimate how the stock's one-minute returns load on contemporaneous and lagged index futures returns and compute the price delay. We relate the change on closure days to the stock's northbound holdings frozen before each episode, with the surrounding trading days as controls. Stocks with larger holdings show a larger rise in delay on closure days. Without controls, their contemporaneous loading and R^2 on index futures also fall. The effect appears with the closure, shows no trend before it, and largely disappears on the day the channel reopens. In the model, the estimate implies that one percentage point of holdings

adds about 2% to the speed of adjustment of a typical stock.

Three tests tie the effect to access rather than to foreign news or foreign ownership. First, many closure days are also holidays in the US and Europe, when news from abroad slows down (Zhang, 2025). The effect is at least as large on the 24 episodes on which only Hong Kong is on holiday as on those shared with Western markets. Second, on 58 US market holidays, when the channel is open, holdings show no effect. Third, holdings of Qualified Foreign Institutional Investors among the top ten shareholders, who trade through mainland brokers that stay open, show no rise in delay. The effect is unchanged when we drop stocks that are also listed in Hong Kong, and 85% of it remains when we control for each stock's exposure to Hong Kong prices.

Foreign-held stocks are larger, more often in the major indices, and better covered by analysts, so we let the closure effect vary with sixteen such characteristics. About a third of the effect on delay remains. Closure days are also somewhat quieter, with lower futures volume. When we add the day's futures variance and market trading value, each interacted with holdings, more than 90% of the controlled effect on delay remains, but the decline in R^2 does not. We therefore rest our conclusions on delay. Days with low futures variance alone also show higher delay for stocks with larger holdings, so we cannot exclude that part of the closure effect reflects calmer markets on closure days. Spreads do not widen.

The study makes three contributions. First, it shows that the information role of foreign investors depends on their access to the market, day by day, and not only on their ownership. Prior work relates foreign ownership and market opening to price efficiency (Bae et al., 2012; Kacperczyk et al., 2021; Lei and Lu, 2024). We hold ownership fixed and switch off access, and we find that the prices of foreign-held stocks absorb market news more slowly as soon as foreign owners cannot trade. Second, it gives price delay a structural interpretation. In our model, delay measures the speed at which the capital present in a stock

incorporates market-wide news, and a change in the capital that can trade maps into a change in delay. This mapping turns a closure into a measure of how much speed of adjustment a class of investors supplies. Third, it adds to work on market closures and the absence of traders. Information that accumulates during halts is impounded once trading resumes (Corwin and Lipson, 2000; Barclay and Hendershott, 2003), and foreign holidays reduce market quality when information from abroad stops (Zhang, 2025). Arbitrage capital moves slowly (Duffie, 2010), arbitrage between index futures and stocks is central to price formation (Hasbrouck, 2003), and liquidity supports efficiency (Chordia et al., 2008). We show that removing one class of traders, while the market and its news flow continue, slows the absorption of market news without widening spreads, and that other traders do not take over within the day.

The remainder of this study is organized as follows. Section 2 describes the closures and develops the model, Section 3 describes the data, Section 4 presents the empirical design, Section 5 reports the results, and Section 6 concludes.

2. Institutional Background and Model

2.1. Stock Connect closures

Stock Connect lets investors in Hong Kong trade eligible Shanghai and Shenzhen stocks through the Hong Kong exchange. This northbound channel opens only on days on which both markets are open and settlement services are available in both places. On Hong Kong public holidays that are not mainland holidays, such as Good Friday, Easter Monday, Buddha's Birthday, the day after the Mid-Autumn Festival, and Christmas, the mainland exchanges trade and the index futures market in Shanghai is open, but investors who hold A-shares through Stock Connect cannot buy or sell. Some of these days are also holidays in the US and Europe, and others are not. Holdings through the channel are disclosed daily by the Hong Kong exchange until August 2024 and quarterly afterward. Foreign institutions can

also trade A-shares through the Qualified Foreign Institutional Investor program, which uses mainland brokers and stays open on these days.

2.2. A partial-adjustment model of price delay

We model how the price of stock *i* absorbs market-wide news within a trading day *t*, with time measured in minutes τ. The efficient value $v_{i,\tau}$ of the stock moves with the index futures return f_τ, which we take as the arrival of market-wide news, and with firm-specific news $\eta_{i,\tau}$:

$$v_{i,\tau}-v_{i,\tau-1}=\beta_i f_\tau+\eta_{i,\tau}, \quad (1)$$

where β_i is the stock's exposure to the market. The futures market absorbs news quickly, so we treat f_τ as serially uncorrelated with variance σ_f^2, and $\eta_{i,\tau}$ as uncorrelated with the futures returns at all leads and lags.

The price $p_{i,\tau}$ does not move to the efficient value at once. Traders who act on market-wide news buy or sell the stock when the futures price moves, and their trades close a fraction $\lambda_{i,t}$ of the remaining gap each minute:

$$p_{i,\tau}-p_{i,\tau-1}=\lambda_{i,t}(v_{i,\tau}-p_{i,\tau-1}), \quad (2)$$

where $\lambda_{i,t}$ lies between zero and one and is the speed of adjustment of the stock on day *t*. Dropping the subscripts of β and λ, the stock's one-minute return is then a distributed lag of futures returns:

$$r_{i,\tau}=\beta\lambda[f_\tau+(1-\lambda)f_{\tau-1}+(1-\lambda)^2 f_{\tau-2}+\ldots]+u_{i,\tau}, \quad (3)$$

where $u_{i,\tau}$ collects the response to firm-specific news and is uncorrelated with the futures returns. The loading on the futures return *k* minutes earlier is

$$b_k=\beta\lambda(1-\lambda)^k,\ k=0,1,2,\ldots, \tag{4}$$

The empirical measures follow directly. *Sync*, the R^2 of the stock return on the contemporaneous futures return alone, equals $b_0^2\sigma_f^2$ divided by the variance of the stock return. Because the futures returns are serially uncorrelated, the R^2 with five lags equals the sum of $b_k^2\sigma_f^2$ for *k* from zero to five, divided by the same variance. The price delay, one minus the ratio of the two, is therefore

$$\text{Delay}=1-1/[1+(1-\lambda)^2+(1-\lambda)^4+\ldots+(1-\lambda)^{10}], \tag{5}$$

Two properties of Equation (5) guide the empirical work. First, delay depends only on the speed of adjustment. It does not depend on the stock's exposure β, on the volatility of the market, or on firm-specific news, so it compares stocks of different size and risk on the same scale. *Sync* also rises with λ, but it falls when firm-specific news becomes more volatile: with *s* equal to the variance of $u_{i,\tau}$ divided by $\beta^2\sigma_f^2$, *Sync* equals $\lambda^2(2-\lambda)/[\lambda+s(2-\lambda)]$. Second, delay falls as λ rises. The mean loadings in our sample, 0.47 on the contemporaneous futures return and 0.36 on the five lags combined, imply through Equation (4) a speed of adjustment of about 0.56, so a typical stock absorbs about half of a futures move in the first minute. The mean measured delay, 60%, is far above the 19% that Equation (5) gives at this speed. With about 240 minutes per day and a contemporaneous R^2 of about 4%, the five lags add R^2 by chance, so the measured delay contains a large estimation component. We therefore use only its change on closure days relative to control days of the same stock.

The speed of adjustment rises with the capital that trades the stock on market-wide news. Let $Hold_i$ be the share of the stock held through northbound Stock Connect, $QFII_i$ the share held by foreign investors with other access, and $Closed_t$ equal one on a day on which the channel is closed. We assume

$$\lambda_{i,t}=\lambda_0+\kappa(1-Closed_t)Hold_i+\kappa_Q QFII_i, \quad (6)$$

where λ_0 is the speed delivered by other traders, $\kappa>0$, and $\kappa_Q\geq 0$. Holdings proxy for the capital that the owners deploy in the stock, and they raise the speed of adjustment only on days on which the owners can trade. Northbound holders can trade only when the channel is open, whereas holders with other access can trade on every mainland trading day.

2.3. Propositions

Proposition 1. On a closure day, the speed of adjustment of stock *i* falls by $\kappa Hold_i$. Its contemporaneous loading on the futures return falls, its price delay rises, and its *Sync* falls, and the rise in delay increases with $Hold_i$. For a small change, the rise in delay is approximately $\kappa Hold_i$ times the absolute value of the slope of Equation (5) in λ.

The proposition follows because Equation (5) decreases in λ and *Sync* increases in λ between zero and one. It gives the main prediction: on closure days, the delay of stocks with larger northbound holdings rises more. The model ranks the two measures. A closure changes delay only through λ, but it can change *Sync* also through the volatility of firm-specific news relative to the market, so delay is the cleaner test.

Proposition 2. The share of a futures move that reaches the price within *H* minutes is $1-(1-\lambda)^{H+1}$. On a closure day, this share falls for every *H*, and the shortfall shrinks as *H* grows. The sum of the loadings on the first to the fifth lag, $\beta[(1-\lambda)-(1-\lambda)^6]$, rises as λ falls when λ is above $1-6^{-1/5}$, about 0.30, but by less than the contemporaneous loading falls.

The proposition describes where the missing part of the response goes. At λ near 0.56, a closure moves part of the response from the first minute to the following minutes, so the lagged loadings rise, but the share that reaches the price within five minutes still falls. The model therefore predicts weaker comovement at short horizons that fades as the horizon lengthens.

Proposition 3. The closure effect requires the channel to close and runs only through northbound holdings. On days on which foreign markets are on holiday but the channel is open, $Closed_t$ is zero and the speed of adjustment does not change. Holdings of foreign investors with other access do not interact with the closure.

The proposition separates access from ownership and from foreign news. If the effect came from foreign holidays reducing the news that reaches China, it would appear on US holidays with the channel open and would not be larger when only Hong Kong is closed. If it came from foreign ownership in general, it would appear for QFII holdings as well.

The model concerns how fast common news is absorbed, not the cost of immediate trading. If northbound investors mainly trade on market-wide news rather than supplying liquidity, a closure lowers trading value by their share of trading but need not widen spreads. Equation (5) is also a population relation. Because the lags add R^2 by chance, measured delay rises whenever the contemporaneous R^2 falls, whether the cause is a lower speed of adjustment or a calmer futures market that leaves firm-specific news a larger share of the stock's variance. Lower market activity on closure days could therefore raise measured delay even with an unchanged speed, and we control for the day's market variance and trading value in Section 5.8.

3. Data

3.1. Sample

We take the northbound trading calendar from the annual Stock Connect calendars of

the Hong Kong exchange and keep days on which northbound trading is closed for the whole day while the Shanghai and Shenzhen exchanges are open. Between January 2017 and October 2026, there are 92 such days in 46 episodes, which last one to three trading days. For each episode, the control days are the five trading days before and the five trading days after it on which northbound trading is open. The sample covers all A-shares listed for at least 60 trading days, excluding special-treatment stocks and stock-days with fewer than 180 one-minute bars or a single price for the whole day.[1] We build the measures from 2.5 billion stock-minutes of one-minute bars. For the futures measures, we use the CSI 300 (IF) and SSE 50 (IH) index futures contracts with the largest volume on the previous trading day. Futures data end in September 2026, so the estimation sample has 45 episodes and 2,207,616 stock-days from 5,409 stocks and 537 trading days.

3.2. Variables

We aggregate the minute data into one number per stock and day, so the unit of analysis is a stock-day. For each stock-day, we regress the stock's one-minute return on the contemporaneous return of the futures contract and five lags, the empirical counterpart of Equation (3):

$$r_{i,\tau}=a+b_0 f_\tau+b_1 f_{\tau-1}+\ldots+b_5 f_{\tau-5}+e_{i,\tau}, \quad (7)$$

where $r_{i,\tau}$ is the stock's one-minute log return and f_τ is the log return of the futures contract. *Beta*0 is one hundred times b_0, and *LagIF* is one hundred times the sum of b_1 to b_5. *Sync* is one hundred times the R^2 of the regression without the lags, which measures how much of the stock's minute-to-minute variation moves with the futures market. We define *DelayIF* as one

1 Minute bars, northbound and QFII holdings, and futures prices are obtained from QDC, a commercial data vendor that redistributes exchange data. The closure calendar is taken from the Stock Connect calendars published by the Hong Kong exchange.

minus the ratio of *Sync* to the R^2 with the lags, following Hou and Moskowitz (2005), expressed in percent; it is the empirical counterpart of Equation (5). *DelayIH* is the same measure with the SSE 50 contract. *Hold* is the percentage of the stock's listed A-shares held through northbound Stock Connect, measured on the last disclosure date before the first control day of each episode and held fixed within the episode. After August 2024, we use the latest quarterly disclosure that is public before the episode. *Hold* is zero for stocks that are not eligible for northbound trading and have no northbound holdings. *Eligible* is one for stocks eligible for northbound buying before the episode. We also use the log trading value (*Value*), the spread estimators of Corwin and Schultz (*SpreadCS*) and Abdi and Ranaldo (*SpreadAR*) computed from minute bars, and the Amihud ratio (*Amihud*). We winsorize every outcome at the 1st and 99th percentiles using thresholds from earlier normal days of the same stock, so that closure days never set the thresholds. *Hold* and *Eligible* are not winsorized.

Table 1. Variable definitions and summary statistics.
Panel A. Definitions

	Definition
Outcome variables	
DelayIF	One hundred times one minus the ratio of the R^2 from a regression of the stock's one-minute returns on the contemporaneous CSI 300 index futures (IF) return to the R^2 when five lagged futures returns are added; estimated each stock-day; in percent
DelayIH	DelayIF computed with the SSE 50 index futures (IH)
Sync	One hundred times the R^2 of the regression of the stock's one-minute returns on the contemporaneous IF return alone
Beta0	One hundred times the coefficient on the contemporaneous IF return in the regression with five lags
LagIF	One hundred times the sum of the five lag coefficients in the same regression
Value	Natural log of daily trading value in yuan
SpreadCS	Daily mean of the spread estimator of Corwin and Schultz from adjacent one-minute bars, in basis points
SpreadAR	Daily mean of the spread estimator of Abdi and Ranaldo from one-minute bars, in basis points
Amihud	Daily mean of the absolute one-minute return divided by one-minute trading value in billion yuan
Explanatory variables	
Closed	One on a day on which northbound Stock Connect is closed for the whole

	day while A-shares trade
Hold	Percentage of listed A-shares held through northbound Stock Connect, frozen before each episode; zero for non-eligible stocks without holdings
Eligible	One if the stock is eligible for northbound buying before the episode
QFII	Percentage of shares held by Qualified Foreign Institutional Investors among the top ten shareholders in the last report before the episode

Panel B. Summary statistics

	Mean	SD	Min	P25	Med	P75	Max
DelayIF	60.03	29.76	1.089	33.79	62.81	88.49	100.00
DelayIH	64.88	28.45	1.309	41.16	70.10	91.64	100.00
LagIF	36.50	61.67	-355.1	1.060	32.21	67.25	782.7
Value	18.23	1.288	11.72	17.31	18.13	19.07	24.24
SpreadCS	6.504	5.926	0.000	3.032	4.916	7.955	544.4
SpreadAR	10.78	8.120	0.000	5.973	9.567	14.20	594.3
Amihud	36.11	253.0	0.007	2.988	10.40	34.21	120,805.5
Hold	0.767	1.818	0.000	0.000	0.030	0.770	34.99
Hold, eligible	1.608	2.351	0.000	0.370	0.840	1.830	34.99
Eligible	0.467	0.499	0.000	0.000	0.000	1.000	1.000
QFII	0.140	0.787	0.000	0.000	0.000	0.000	20.02

Notes. This table defines the variables and reports their summary statistics. In Panel B, each observation is a stock-day in the estimation sample of the closure episodes and their control days. Outcome variables are winsorized at the 1st and 99th percentiles of earlier normal days of the same stock, and *Hold*, *Eligible*, and *QFII* are not winsorized. *Hold, eligible* is *Hold* for eligible stocks. The sample has 2,207,616 stock-days, of which 1,030,034 are of eligible stocks, and *QFII* is available for 2,198,195. *Mean*, *SD*, *Min*, *P*25, *Med*, *P*75, and *Max* are the mean, standard deviation, minimum, 25th percentile, median, 75th percentile, and maximum.

Table 1 defines the variables and reports their summary statistics. The mean of *DelayIF* is 60%, so on a typical stock-day the lags account for a large part of the explained variation in minute returns. *Hold* is zero or close to zero for half of the stock-days, and its median among eligible stocks is 0.84%. Table 2 shows that the episodes are spread evenly over the years and that eligible stocks are larger, trade more, have more analysts, and have lower delay than other stocks. Stock-fixed effects absorb these level differences, but such stocks may also respond differently on closure days, so Table 3 lets the closure effect vary with these characteristics.

Table 2. Sample composition and balance.

Panel A. Episodes by year

	Episodes	Days	Stocks	Eligible	Stock-days
2017	3	8	2,957	1,414	106,826

2018	6	14	3,325	1,408	239,898
2019	4	10	3,482	1,294	173,016
2020	5	11	3,677	1,357	222,308
2021	6	13	4,113	1,452	294,127
2022	4	10	4,533	1,494	224,938
2023	4	7	4,846	2,644	227,142
2024	6	8	4,955	2,827	335,997
2025	4	6	4,975	2,854	227,825
2026	4	5	4,980	3,381	223,383
All	46	92	4,193	1,996	2,275,460

Panel B. Eligible and other stocks

	Eligible	Others	Diff	t	StdDiff
Hold	1.604	0.031	1.572	(24.38)	0.934
Value	18.82	17.71	1.114	(28.82)	1.086
Size	23.18	21.67	1.501	(118.06)	1.682
BetaIF	0.926	0.788	0.138	(5.32)	0.282
PreDelay	0.511	0.675	-0.164	(-21.55)	-0.942
Analysts	1.904	0.691	1.212	(53.05)	1.178

Notes. In Panel A, each row is a calendar year. *Episodes* is the number of closure episodes, *Days* the number of closure days, *Stocks* and *Eligible* the mean numbers of sample stocks and eligible stocks per episode, and *Stock-days* the number of stock-days in the closure and control days. The number of eligible stocks rises in 2023, when the eligibility rules were widened. In Panel B, each observation is a stock in one episode, measured before the episode; there are 87,819 eligible and 99,580 other stock-episodes. *Value* is the log of median daily trading value over the 60 previous normal days, *Size* is the log of float market value, *BetaIF* is the beta of daily returns on IF returns, *PreDelay* is the mean of *DelayIF* divided by 100 over the 60 previous normal days, and *Analysts* is the log of one plus the number of brokers covering the stock over the previous year. *Diff* is the difference in means, *t* is its t-statistic with standard errors clustered by episode, and *StdDiff* is the difference divided by the pooled standard deviation.

4. Empirical Design

Proposition 1 predicts that the delay of stock *i* rises on closure days in proportion to $Hold_i$. We relate each outcome of each stock on each day to the closure indicator interacted with the stock's holdings:

$$Y_{i,t}=\beta(Closed_t\times Hold_{i,e})+\delta Hold_{i,e}+\alpha_i+\mu_t+\varepsilon_{i,t}, \tag{8}$$

where $Y_{i,t}$ is *DelayIF*, *DelayIH*, *Sync*, *Beta*0, or another outcome of stock *i* on day *t*, $Closed_t$ is one on a closure day, and $Hold_{i,e}$ is the holdings of stock *i* frozen before episode *e*. The level of $Hold_{i,e}$ is included because it changes across episodes. α_i and μ_t are stock-fixed effects and

date-fixed effects, so the date-fixed effects absorb all market-wide changes on closure days, and β measures how much more the outcome changes on closure days for each percentage point of holdings. In the model, β for delay equals κ times the slope of Equation (5) in λ. We cluster standard errors by stock and by date. Because the closure shocks come from 45 episodes, we also report wild cluster bootstrap p-values with clustering by episode and 9,999 draws.

5. Results

5.1. Closure days and price delay

Table 3. Closure days, price delay, and comovement with index futures.

	DelayIF	DelayIH	Sync	Beta0	Beta1	LagIF
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Without controls						
Closed×Hold	0.634***	0.631***	-0.226***	-0.975***	0.143	0.416
	(5.85)	(5.92)	(-6.20)	(-4.62)	(0.83)	(1.36)
Wild p	<0.001	<0.001	<0.001	<0.001	0.372	0.168
Episodes	45	45	45	45	45	45
Obs	2,207,616	2,207,616	2,207,616	2,210,311	2,210,311	2,210,311
Panel B. With controls						
Closed×Hold	0.232***	0.228***	-0.067***	-0.340***	0.000	0.056
	(3.85)	(3.57)	(-4.28)	(-3.49)	(-0.00)	(0.36)
Wild p	<0.001	<0.001	<0.001	<0.001	0.998	0.712
Episodes	44	44	44	44	44	44
Obs	2,127,264	2,127,264	2,127,264	2,129,809	2,129,809	2,129,809
Mean	60.03	64.88	4.203	47.23	24.47	36.46
Stock_FE	Yes	Yes	Yes	Yes	Yes	Yes
Date_FE	Yes	Yes	Yes	Yes	Yes	Yes

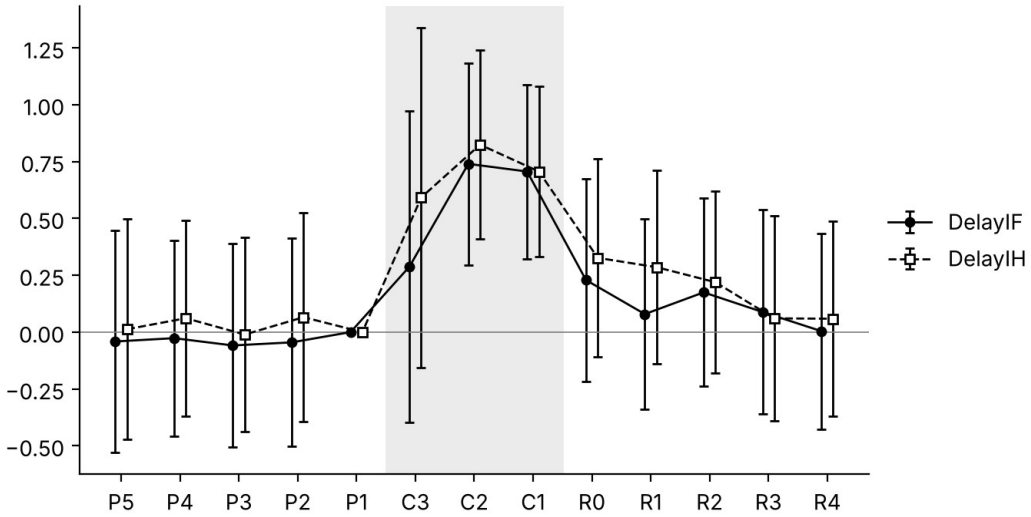
Notes. Each observation is a stock-day in the closure episodes and their control days. *Closed* is one on a closure day, and *Hold* is the percentage of shares held through northbound Stock Connect before the episode, whose level is included but not reported. *Beta*1 is one hundred times the coefficient on the first lag of the IF return. Panel B adds *Closed* interacted with sixteen characteristics measured before the episode: log float market value, log trading value, membership in the CSI 300, SSE 50, and CSI 500 indices, the beta on IF, the pre-closure delay, analyst coverage, and eight measures of exposure to Hong Kong prices; the levels are included but not reported, and one episode drops because index membership data are missing. *Mean* is the sample mean of the dependent variable. *Wild p* is the wild cluster bootstrap p-value with clustering by episode and 9,999 draws. The t-statistics in parentheses are calculated using standard errors clustered by stock and by date. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table 3 reports the main estimates. In Panel A, *DelayIF* and *DelayIH* rise more on

closure days for stocks with larger northbound holdings, and the effect is significant at the 1% level, also under the wild bootstrap. Each percentage point of holdings raises the delay by 0.63 percentage points. Columns (3) to (6) show the other predictions of Propositions 1 and 2. *Sync* falls by 0.23 for each percentage point of holdings, which is 5% of its mean, and the contemporaneous loading *Beta*0 falls by about one point. Through Equation (4), this fall implies that one percentage point of holdings adds about 0.012 to the speed of adjustment, or 2% of its mean of 0.56. The lagged loadings rise by less than half as much as *Beta*0 falls, and not significantly, as Proposition 2 predicts at this speed.

Northbound investors prefer large stocks, index constituents, and stocks covered by analysts, and such stocks may respond differently to market news on any closure day. Panel B therefore lets the closure effect vary with sixteen characteristics measured before the episode, including size, trading value, index membership, futures beta, the pre-closure delay, analyst coverage, and exposure to Hong Kong prices. The estimates fall to about 37% of those in Panel A for the delay measures and to about 30% for *Sync*, and they remain significant at the 1% level. With controls, a one-standard-deviation increase in holdings, 1.8 percentage points, raises the delay on closure days by about 0.4 percentage points, which is modest relative to the day-to-day variation in delay, and the fall in *Beta*0 implies a change in the speed of adjustment of about 0.004 per percentage point of holdings. The step-by-step results show that size accounts for most of the decline, and that the effect is also present within eligible stocks and within size deciles on each date (Online Appendix Table OA5).

Figure 1. Price delay around closure episodes, measured from the reopening day.



Notes. The figure plots the coefficients of *DelayIF* and *DelayIH* on *Hold* interacted with indicators for trading days relative to the day northbound trading reopens, with the last normal day before the closure as the reference, and their 95% confidence intervals. On the horizontal axis, *P*5 to *P*2 are the normal days before the closure, *C*3 to *C*1 are closure days counted back from the reopening, and *R*0 to *R*4 are the reopening day and the following days; *C*3 exists in 11 episodes and *C*2 in 35. The vertical axis is in percentage points of delay. The estimation uses 45 episodes with stock-fixed effects and date-fixed effects, and the confidence intervals use standard errors clustered by stock and by date.

Figure 1 traces the effect around each episode, with the event time measured from the day the channel reopens. The estimates are close to zero on each of the days before the closure. They jump on the closure days and fall to between one-third and one-half of that size, which is not significant, on the reopening day. They are small and not significant afterward. With the sixteen controls, the estimates before the closure remain close to zero, the closure-day estimates remain positive, and the reopening-day estimate for *DelayIF* is not significant. The pattern follows the closure rather than a gradual change around holidays.

5.2. Comovement at longer horizons

Proposition 2 predicts that the missing part of the response arrives more than five minutes later and that the shortfall fades as the horizon lengthens. Table 4 measures the stock's beta and R^2 on index futures at frequencies from one minute to one hour, and the stock's daily and next-morning response.

Table 4. Comovement at longer horizons.

	1Min	5Min	15Min	30Min	60Min	Day	Next
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Beta on IF							
Treatment	-1.046***	-0.802**	-1.405**	-0.536	-3.867***	1.327	2.198**
	(-4.52)	(-2.21)	(-2.57)	(-0.70)	(-2.60)	(0.91)	(2.18)
Wild p	<0.001	0.004	0.004	0.405	0.004	0.338	0.027
Mean	45.90	69.63	74.21	77.27	72.63		
Panel B. R^2 on IF							
Treatment	-0.198***	-0.251***	-0.384***	-0.239***	-0.234**		
	(-5.57)	(-5.14)	(-4.89)	(-2.83)	(-2.12)		
Wild p	<0.001	<0.001	<0.001	0.004	0.013		
Mean	4.151	12.01	19.75	27.29	42.53		
Stock_FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date_FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Episodes	45	45	45	45	45	45	45
Obs	2,209,937	2,209,937	2,209,937	2,209,937	2,209,937	2,210,311	2,207,080

Notes. Each observation is a stock-day in the closure episodes and their control days. In columns (1) to (5), the dependent variable is one hundred times the beta or the R^2 of the stock's returns on contemporaneous IF returns at one-, five-, 15-, 30-, and 60-minute frequency, estimated each stock-day without lags, and *Treatment* is *Closed×Hold*. In column (6), *Day*, the dependent variable is the stock's open-to-close return, and *Treatment* is *Closed×Hold* interacted with the IF open-to-close return, so the coefficient is one hundred times the change in the daily beta. In column (7), *Next*, the dependent variable is the stock's return from 9:30 to 10:00 on the next trading day, and *Treatment* is the same interaction with the IF return of the current day. All lower-order terms are included but not reported. *Mean* is the mean on control days. *Wild p* is the wild cluster bootstrap p-value with clustering by episode and 9,999 draws. The t-statistics in parentheses are calculated using standard errors clustered by stock and by date. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

At every frequency from one minute to one hour, the R^2 on index futures falls on closure days for stocks with larger holdings, and the betas fall at all frequencies except 30 minutes. The decline in the one-hour beta is no smaller than that in the one-minute beta, so, contrary to the gradual fading in Proposition 2, the shortfall does not shrink within the hour. Estimating the 30- and 60-minute beta and R^2 once for each stock and episode, pooling the closure days and pooling the control days, gives significant declines as well, although the 30-minute beta is not significant without winsorization. The daily beta does not differ significantly, which leaves open how much of the gap closes by the end of the day. On the next morning, the first half hour of trading moves with the futures return of the closure day for stocks with larger holdings, with a wild p-value of 0.03 that rises to 0.06 when we adjust

for testing both the daily and the next-morning response. Foreign-held stocks therefore follow the market less closely for at least an hour on closure days, and part of the missing move may arrive when trading resumes. Relative to the model, the speed of adjustment appears to fall by more at longer horizons than a constant λ implies, as it would if the traders who remain also respond more slowly through the day.

5.3. Foreign news or foreign access

Many closure days are also holidays in the US or Europe, when less news arrives from abroad. Lower comovement on these days may then reflect less foreign news rather than the closed channel. A second possibility is that foreign holidays change trading by foreign investors, whether or not the channel is open. Proposition 3 rules out both channels in the model, and Table 5 tests them.

Table 5. Foreign news or foreign access.

	HKOnly		Shared		USHoliday	
	DelayIF	DelayIH	DelayIF	DelayIH	DelayIF	DelayIH
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Without controls						
Event×Hold	0.683***	0.655***	0.523***	0.547***	-0.007	0.005
	(5.53)	(5.44)	(3.13)	(3.36)	(-0.05)	(0.04)
Gap			0.178	0.126	-0.561***	-0.561***
			(0.87)	(0.63)	(-3.44)	(-3.56)
Wild p	<0.001	<0.001	0.008	0.010	0.949	0.971
Wild p, Gap			0.366	0.556	0.003	0.002
Episodes	24	24	21	21	58	58
Obs	1,140,995	1,140,995	1,066,621	1,066,621	2,573,660	2,573,660
Panel B. With controls						
Event×Hold	0.344***	0.303***	0.146**	0.165**	0.050	0.061
	(3.55)	(2.91)	(2.18)	(2.38)	(0.74)	(0.89)
Gap			0.230	0.190		
			(1.33)	(1.01)		
Wild p	<0.001	0.005	0.060	0.026	0.305	0.258
Wild p, Gap			0.231	0.368		
Episodes	23	23	21	21	55	55
Obs	1,077,520	1,077,520	1,049,744	1,049,744	2,440,603	2,440,603
Stock_FE	Yes	Yes	Yes	Yes	Yes	Yes
Date_FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Each observation is a stock-day in the event days and their control days. In columns (1) and (2), *HKOnly*, the events are closure episodes on which the US, UK, and euro area markets are open, and in columns (3) and (4), *Shared*, closure episodes on which at least one

of these markets is also closed; *Event* is *Closed*. In columns (5) and (6), *USHoliday*, the events are US market holidays on which northbound trading is open, *Episodes* counts these holidays, and *Event* is one on such a holiday. *Gap* in columns (3) and (4) is the coefficient for *HKOnly* minus that for *Shared*, and in columns (5) and (6) of Panel A the holiday coefficient minus the closure coefficient, each estimated in one regression that includes both groups of events; in Panel B, the gap regression uses common slopes for the controls. Panel B adds *Closed* interacted with sixteen characteristics measured before the episode: size, trading value, index membership, the beta on IF, the pre-closure delay, analyst coverage, and eight measures of exposure to Hong Kong prices. The level of *Hold* is included but not reported. *Wild p* is the wild cluster bootstrap p-value with clustering by episode and 9,999 draws. The t-statistics in parentheses are calculated using standard errors clustered by stock and by date. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

In columns (1) and (2), we use the 24 episodes on which the US, UK, and euro area markets are open. The effect is at least as large as in the full sample, and with controls its point estimate is larger than on the 21 episodes shared with at least one Western market, although the difference is not significant. With controls, one HK-only episode drops because the CSI 500 member list before it is incomplete, and three US holidays drop because of missing history or index data. In columns (5) and (6), we use 58 US market holidays on which northbound trading is open. Holdings show no effect with or without controls, and the difference from the closure effect is significant at the 1% level. The effect therefore requires the channel to close, and it is not weaker when foreign markets are open.

5.4. Hong Kong prices and foreign investors outside Stock Connect

Hong Kong prices stop on closure days, so stocks held by northbound investors may lose a source of price signals, including their own H-shares. The effect may also reflect foreign ownership in general rather than the channel.

Table 6. Hong Kong prices and foreign investors outside Stock Connect.

	NoAH		HKExp		QFII	
	DelayIF	DelayIH	DelayIF	DelayIH	DelayIF	DelayIH
	(1)	(2)	(3)	(4)	(5)	(6)
Closed×Hold	0.641***	0.616***	0.538***	0.566***	0.641***	0.634***
	(5.97)	(5.73)	(6.17)	(6.20)	(6.00)	(5.97)
Closed×QFII					-0.029	0.003
					(-0.39)	(0.04)
Wild p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
MDE, QFII					0.213	0.243

Stock_FE	Yes	Yes	Yes	Yes	Yes	Yes
Date_FE	Yes	Yes	Yes	Yes	Yes	Yes
Episodes	45	45	45	45	45	45
Obs	2,137,668	2,137,668	2,173,271	2,173,271	2,198,195	2,198,195
R^2	0.006	0.005	0.070	0.046	0.006	0.005

Notes. Each observation is a stock-day in the closure episodes and their control days. Columns (1) and (2), *NoAH*, exclude stocks also listed in Hong Kong. Columns (3) and (4), *HKExp*, add *Closed* interacted with eight measures of the stock's exposure to Hong Kong prices before the episode: the beta, correlation, and incremental R^2 of its one-minute returns on Hong Kong index exchange-traded funds listed in Shanghai and Shenzhen, and the betas and correlations of its daily returns on the Hang Seng Index and the Hang Seng China Enterprises Index. Columns (5) and (6), *QFII*, add *Closed*×*QFII*. *MDE, QFII* is the minimum detectable effect of *Closed*×*QFII* at 5% significance and 80% power. The levels of the interacted variables are included but not reported. *Wild p* is the wild cluster bootstrap p-value with clustering by episode and 9,999 draws. The t-statistics in parentheses are calculated using standard errors clustered by stock and by date. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table 6 addresses both alternatives. Dropping the stocks that are also listed in Hong Kong leaves the estimate unchanged. Interacting the closure indicator with eight measures of each stock's exposure to Hong Kong prices, built from Hong Kong index funds traded in Shanghai and Shenzhen and from daily Hang Seng index returns, retains about 85% of the estimate. Holdings of Qualified Foreign Institutional Investors among the top ten shareholders show no closure effect on the delay, whereas the coefficient on northbound holdings is unchanged. The minimum detectable effect for these holdings is about one-third of the northbound effect without controls. With the sixteen controls, the coefficient on QFII holdings for *DelayIF* is −0.08 and its minimum detectable effect is 0.18, about three-quarters of the controlled northbound estimate, so this comparison has less power. With controls, QFII holdings also show a small rise in *Sync* on closure days, the opposite sign of the northbound estimate. These results are consistent with an effect that runs through the channel that closes. Tables 6 to 8 and Figure 1 report estimates without controls unless stated otherwise.

5.5. Evidence by episode

The panel estimates pool more than two million stock-days, but the closure shocks come from 45 episodes. Table 7 therefore reduces each episode to one number.

Table 7. Evidence by episode.

	Quintile		Matched	
	DelayIF	DelayIH	DelayIF	DelayIH
	(1)	(2)	(3)	(4)
Difference	4.178***	3.768***	2.015***	1.759***
	(7.02)	(5.52)	(4.90)	(4.34)
Difference, controls	2.095***	1.565**		
	(3.56)	(2.30)		
Positive	41	40	35	29
Episodes	45	45	45	45
Sign p	<0.001	<0.001	<0.001	0.072

Notes. Each observation is an episode. For each stock, we compute the mean delay on closure days minus the mean on control days. In columns (1) and (2), *Quintile*, *Difference* is the mean across episodes of the difference between the top and bottom quintiles of *Hold* within the episode. In columns (3) and (4), *Matched*, it is the mean across episodes of the difference between stocks in the top quartile of positive *Hold* and matched stocks with low or no holdings in the same industry and of similar size, trading value, and IF beta. *Difference, controls* is the quintile difference after removing the closure effects of the sixteen characteristics of Table 3, Panel B, estimated with common slopes, in 44 episodes. *Positive* is the number of episodes with a positive difference in the first row, and *Sign p* is the p-value of the exact sign test. The t-statistics in parentheses are calculated across episodes. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

In columns (1) and (2), we sort stocks into quintiles of *Hold* within each episode and compute the difference between the top and bottom quintiles in the change in delay from control days to closure days. The average difference is 4.2 percentage points for *DelayIF*, and it is positive in 41 of 45 episodes. After removing the closure effects of the sixteen characteristics, the difference is 2.1 percentage points for *DelayIF* and 1.6 for *DelayIH*, both significant at the 5% level across episodes. In columns (3) and (4), we match each stock in the top quartile of positive holdings to the stock with low or no holdings in the same industry that is closest in size, trading value, and futures beta. The matched difference is about half as large, and it is positive in 35 of 45 episodes for IF and in 29 for IH, where the sign test is significant only at the 10% level. The effect is spread across episodes, and about half of the unmatched difference reflects characteristics that differ between high-holding and low-holding stocks.

5.6. Trading activity and liquidity

Table 8. Trading activity and liquidity.

	Value	SpreadCS	SpreadAR	Amihud
	(1)	(2)	(3)	(4)
Closed×Hold	-0.012***	0.019	-0.003	-0.282
	(-3.08)	(1.04)	(-0.11)	(-0.85)
Closed×Eligible	-0.053***	-0.093	-0.300***	-1.431
	(-2.72)	(-1.36)	(-3.19)	(-0.67)
Closed×Hold, controls	-0.008***	-0.001	0.004	0.132
	(-4.52)	(-0.11)	(0.30)	(0.66)
Wild p, Hold	<0.001	0.038	0.869	0.208
Wild p, Eligible	<0.001	0.027	<0.001	0.288
Wild p, controls	<0.001	0.879	0.715	0.203
Mean	18.24	6.516	10.76	35.61
Stock_FE	Yes	Yes	Yes	Yes
Date_FE	Yes	Yes	Yes	Yes
Episodes	46	46	46	46
Obs	2,269,977	2,269,977	2,269,977	2,269,977

Notes. Each observation is a stock-day in the closure episodes and their control days, including the October 2026 episode, for which futures data are not available. *Closed×Hold* and *Closed×Eligible* are estimated in separate regressions that include the level of *Hold* or *Eligible*. *Closed×Hold, controls* adds *Closed* interacted with the sixteen characteristics of Table 3, Panel B, and uses the 45 episodes with futures data and complete controls (2,189,241 stock-days). The sample with *Eligible* has 5,483 more stock-days than *Obs*, because these stock-days lack holdings data. *Mean* is the sample mean of the dependent variable. *Wild p, Hold*, *Wild p, Eligible*, and *Wild p, controls* are wild cluster bootstrap p-values with clustering by episode and 9,999 draws. The t-statistics in parentheses are calculated using standard errors clustered by stock and by date. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

If northbound investors provide liquidity, their absence should widen spreads and raise price impact. Table 8 shows that trading value falls by 1.2% for each percentage point of holdings, but the spread estimators and the Amihud ratio show no economically meaningful change. The estimator of Corwin and Schultz rises by 0.02 basis points per percentage point of holdings, which is 0.3% of its mean and significant only under the wild bootstrap, and the estimator of Abdi and Ranaldo falls for eligible stocks. With the sixteen controls, trading value still falls by 0.8% per percentage point, and none of the spread measures or the Amihud ratio changes. As the model allows, the closure slows the absorption of market news without a comparable change in the cost of trading.

5.7. Robustness

The results hold when we use the eligibility indicator, the log of holdings, or the top quintile of holdings as the treatment, when we drop the largest holdings, when we change the winsorization, and when we measure the delay with five-minute returns (Online Appendix Table OA1). Normal days next to holidays that the mainland and Hong Kong share, and dates shifted back by 20 or 40 trading days, show no effect (Online Appendix Table OA2). The holiday-adjacent placebo has 14 episodes and wider confidence intervals, so it cannot exclude small effects. Before August 2024, quarter-end holdings give the same estimate as daily holdings, so the change to quarterly disclosure does not alter the treatment (Online Appendix Table OA3). The estimate for the ten later episodes is smaller and not significant, but its minimum detectable effect exceeds the full-sample estimate. Six full-day closures caused by typhoon or rainstorm warnings, which are not on the annual calendar, give estimates of similar size, with wild p-values of 0.07 for *DelayIF* and 0.04 for *DelayIH*; six half-day closures give 0.05 for *DelayIF* (Online Appendix Table OA4). Stocks that permanently lose northbound buying after index reviews show a larger rise in delay than stocks that leave the same indices but remain buyable in 10 of the 13 batches that can be estimated, but the estimate is imprecise.

5.8. Market activity on closure days

Closure days are somewhat quieter than their control days. Volume in IF futures is 7% to 8% lower, which is significant, and market trading value and IF realized variance are 7% and 10% lower, which is not significant. As noted in Section 2.3, a weaker market signal can raise measured delay. If the measured delay of high-holding stocks is more sensitive to a weak signal, the closure estimate could reflect calmer markets rather than missing traders.

We address this in two ways (Online Appendix Table OA6). First, we add the day's IF realized variance and market trading value, each interacted with *Hold* and with *Closed*×*Hold*. Without other controls, 96% of the *DelayIF* estimate and 91% of the *DelayIH* estimate

remain. With the sixteen characteristics, 93% and 79% remain, significant at the 5% level under the wild bootstrap. The *Sync* estimate with the sixteen characteristics, however, falls to zero. Controlling for the number of minutes with nonzero returns retains 80% of the delay estimate. Second, we treat normal days with low market activity as placebo events. On days on which both market trading value and IF variance are in the bottom 30% or 40% of their earlier distribution, 34 and 85 days, stocks with larger holdings show no significant rise in delay, and for the 40% definition, the closure effect is significantly larger than the quiet-day effect for *DelayIF* and *Sync*. On days defined by low IF variance alone, about 550 to 750 days, stocks with larger holdings show a rise in delay as large as on closure days, and the two cannot be distinguished.

These results show that the closure effect on delay is not explained by the level of market activity on the day, but that delay rises for high-holding stocks whenever the futures market is calm. We therefore cannot rule out that part of the closure effect reflects calmer futures markets on closure days. The US market holidays in Table 5, which also bring less global news, show no effect.

6. Conclusion

We use the scheduled closures of northbound Stock Connect on Hong Kong holidays to switch off foreign access to Chinese stocks for a day or two while ownership stays the same and the mainland market keeps trading. A partial-adjustment model links the price delay of a stock to the capital that trades it on market-wide news and predicts that a closure raises delay in proportion to northbound holdings. On closure days, stocks with larger northbound holdings show a larger rise in delay, and weaker comovement with index futures lasts for at least an hour. The effect is at least as large when only Hong Kong is on holiday, it is absent on US holidays when the channel is open, and it is unrelated to holdings of foreign investors whose access stays open. Spreads do not widen. The information role of foreign

investors in an emerging market therefore depends on their access at each moment, not only on their presence among the owners. For regulators, this means that the calendar of a cross-border channel shapes how fast local prices absorb market news. The closure shocks come from 45 short episodes, about two-thirds of the raw effect reflects the characteristics of foreign-held stocks, and delay also rises for these stocks on calm days, so the estimates measure the loss from short interruptions of access rather than from a permanent closure.

Declaration of generative AI use

During the preparation of this work, the authors used Claude (Anthropic) to draft and edit the text and to check the consistency of the tables with the estimation output, and Codex (OpenAI) to run the statistical code and verify bibliographic records. The authors reviewed and edited all content and take full responsibility for it.

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